

East Asian Authoritarian Developmentalism in the Digital Era

*China's Techno-developmental State and the New
Infrastructure Initiative amid Great Power Competition*

ABSTRACT

This article examines the evolution of the authoritarian developmental state model in the digital era amid intensified global power dynamics, focusing empirically on China. It introduces the concept of the techno-developmental state to illustrate how this evolved state emphasizes digital technologies for both economic growth and social governance. I assert that this evolution is driven by the state's enhanced capacity to foster high-tech industries amid heightened competition among major powers, alongside bolstered social governance through advanced digital tools. Emphasizing its crucial role in sustaining authoritarian developmentalism, the article analyzes the emergence of the Chinese techno-developmental state amid escalating US–China technological rivalry. It also scrutinizes the New Infrastructure initiative of the early 2020s as the state's key initiative to drive its economic and governance agendas.

KEYWORDS: Techno-developmental state, authoritarian developmentalism, digital infrastructure, digital governance, China

DURING THE POSTWAR ERA, East Asian authoritarian developmental states engineered rapid economic progress (Amsden 1989; Haggard 2018; Wade 1990; Woo-Cumings 1999). The 1990s saw a shift toward democracy, leading

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to the decline of authoritarian developmentalism in South Korea and Taiwan (Minns 2001; Tan 2008; Wu 2007). However, in countries like Singapore and China, where state dominance remains pervasive, developmentalism has persisted, maintaining its legitimacy through effective performance (He 2021; Pereira 2008). The developmental state model that emerged in the postwar period is not static but has shown adaptability over the course of development (Wong 2004). With the onset of the digital era and heightened great power competition since the mid-2010s, how have authoritarian developmental states adapted to this new landscape?

Studies have highlighted the emergence of security-driven, tech-focused states in recent times, particularly in the context of China. Lei (2023) examines “techno-developmental regimes,” prioritizing science and technology for socioeconomic progress in the digital era. Cheung (2022) introduces the term “techno-security state” to describe states adept at integrating security, innovation, and development. Drawing on these insights, I explore the rise of the techno-developmental states that strategically deploy technologies not only for economic advancement and security but also to consolidate political power in authoritarian developmental Asia. I argue that this phenomenon significantly perpetuates authoritarian developmentalism in East Asia. These states harness advanced technological capabilities across key sectors to enhance national economic potential and exert control over societal dynamics. This ascent reflects two core strategies embraced by authoritarian developmental states amid evolving geopolitical and social landscapes: enhancing state capacities to oversee critical technological sectors in a competitive geopolitical environment; and integrating digital technologies into social governance frameworks to improve governance efficacy in the digital age.

The article explores the emergence of the techno-developmental state in China. While debates continue regarding China’s adherence to the East Asian developmental state model (Breslin 1996; Howell 2006; Knight 2014), the Chinese authoritarian regime has undeniably been instrumental in driving rapid industrial progress since the introduction of market reforms in the 1980s. Since the late 2010s, China’s developmental path has increasingly emphasized high technology due to escalating geo-strategic competition with the US (Cheung 2022; He and You 2023; Lei 2023). As the Chinese state intensifies its industrial policies to promote high-tech economic growth, the use of emerging technologies for social governance has

become increasingly prominent. The deployment of digital technologies has proven effective in enhancing public services, managing citizen behavior, addressing social grievances (Göbel and Li 2021; Hou and Fu 2022; Xia, Yan, and Yang 2022), promoting social stability (Hu and Zhang 2023; Schlæger and Stepan 2017), monitoring public officials (Schlæger and Wang 2017), and exerting political control (Cabestan 2020; Lin, Wang, and Tsai 2022).

I argue that the Chinese regime's ability to maintain authoritarian developmentalism has been greatly enhanced by the emergence of the techno-developmental state, particularly since the ramp-up of the US–China strategic rivalry. To illustrate how this newly formed techno-developmental state has employed digital technologies to pursue both economic development agendas and social governance objectives, I conduct a detailed analysis of China's recent efforts to advance digital infrastructure through the New Infrastructure (*xin jijian*) initiative in the early 2020s.

In the following sections, I first review the literature that establishes the conceptual framework for the techno-developmental state. I then examine the ascent of this state in China, emphasizing the state's enhanced efforts to incorporate technologies into its governance strategies and bolster its technological capabilities. Furthermore, I analyze how this new techno-developmental state has pursued growth and social governance agendas enabled by technology, with a particular focus on the New Infrastructure initiative. The concluding section summarizes the main argument and explores broader implications for understanding the techno-developmental state's influence in China and globally.

HOW EAST ASIAN AUTHORITARIAN DEVELOPMENTALISM EVOLVES IN THE ERA OF DIGITALIZATION AND GREAT POWER COMPETITION

The Postwar East Asian Authoritarian Developmental State Model

The literature on the developmental state model offers insights into East Asian authoritarian developmentalism in the postwar era. Originating from the imperative of ruling elites to ensure regime survival amid internal and external challenges (Castells 1992; Doner, Ritchie, and Slater 2005), the developmental state model underscores the dual strategy of political control and economic promotion. This approach was crucial for establishing

performance legitimacy and sustaining regimes in East Asia (Castells 1992). While political control allowed elites to maintain authority despite limited popular support, it was insufficient for long-term power consolidation. Therefore, elites also implemented economic policies to foster industrial growth to meet the material expectations of politically discontented segments of society (Doner, Ritchie, and Slater 2005).

The pursuit of this survival strategy necessitates two specific types of political institutions: a policymaking agency and an authoritarian political system (Castells 1992). The policymaking agency plays a pivotal role in driving the development process. By overseeing a competent and disciplined economic bureaucracy composed of skilled managers, such an agency can effectively manage a nation's economic development program in collaboration with the private sector (Amsden 1989; Evans 1995; Johnson 1982; Wade 1990). Examples of these agencies include Taiwan's Council for Economic Planning and Development and South Korea's Economic Planning Board. Authoritarian political institutions enhance the efficacy of state policymaking by insulating the bureaucracy from special interests, thereby providing them with sufficient protection and authority to initiate policies and operate efficiently (Johnson 1987; Wade 1990). This arrangement fosters a situation where, as Chalmers Johnson (1982) describes it, politicians "reign" while bureaucrats "rule." Consequently, these competent economic bureaucrats could withstand rent-seeking pressures from the private sector and maintain a long-term perspective on the development process (Onis 1991).

Merely having an autonomous state isn't sufficient for effective policymaking. As Linda Weiss (1998, 48) observes, "What makes policies effective is a specific state structure and a particular relationship between the state and industry." In East Asia, the autonomous state harnesses its strength and independence by cultivating a cooperative, state-led partnership with the private sector to enhance the effectiveness of its policy interventions (Chibber 2003; Evans 1995; Weiss 1995).

How does the authoritarian developmental state model adapt in the digital era amid escalating great power dynamics? This evolution involves two concurrent dimensions: externally, the state's reaction to the rise of great power competition; and internally, the state's ability to be resilient in upholding authoritarian developmentalism. These dimensions collectively strengthen the state's capabilities in economic management and social governance.

Great Power Politics and Intensified Economic Statecraft in Technological Sectors

The literature on domestically oriented economic statecraft provides insights into how states strategically cultivate technological sectors. These studies emphasize states' efforts to strengthen technological sectors in response to external pressures from competitors (Thurbon and Weiss 2021; Weiss and Thurbon 2021). They highlight that technology-driven economic statecraft typically emerges from two primary external motivations: geoeconomic and geopolitical. Geoeconomic motivations push toward technological independence to enhance commercial competitiveness, whereas geopolitical motivations prioritize technological superiority to bolster military capabilities. The United States exemplifies this strategy through transformative innovation facilitated by close collaboration between federal agencies and private entities to achieve security-related objectives (Weiss 2014).

East Asian authoritarian developmentalism was deeply intertwined with external geopolitical pressures, which served as catalysts for establishing developmental states and implementing industrial policies to drive rapid economic growth (Doner, Ritchie, and Slater 2005; Woo 1992). The formation and functioning of these states were facilitated by persistent yet relatively low-intensity military pressures from external actors, influencing specific threat perceptions across Northeast Asian nations (Zhu 2002). Subsequently, these states, exemplified by China, have played pivotal roles in shaping strategic technological sectors, where national security concerns significantly influence industrial policy, underscoring the integration of civilian and military sectors as fundamental to China's industrial strategy (Cheung 2008). The escalating intensity of great power competition in the 2010s increased geopolitical pressures and reshaped the external environment, further impacting East Asian authoritarian developmentalism. How did these developmental states adapt to this evolving geopolitical scenario?

Research indicates that the escalating US–China strategic rivalry has heightened security-driven, technology-focused economic statecraft efforts in both countries. This competition has spurred significant economic statecraft efforts by the US government to secure geopolitical interests and maintain leadership in high-tech sectors (Weiss 2021; Weiss and Thurbon 2021). Concurrently, since the mid-2010s, its rivalry with the US has fostered robust techno-security measures in China to bolster technological supremacy

(Cheung 2022). Moreover, geopolitical considerations now emphasize economic security as a priority for the Chinese regime's economic statecraft. This trend has prompted a shift toward a securitized approach in governing domestic technology firms (Pearson, Rithmire, and Tsai 2022). Concerns regarding economic security have notably accelerated China's recent initiatives toward technological self-reliance under the leadership of Xi Jinping (Cheung 2022). Aligned with these findings, this article will examine the state's heightened strategic actions in governing the technological sectors amid escalating external pressures from intensified great power competition.

The State's Resilience and the Rise of Digital Governance

A defining feature of the new East Asian developmental authoritarianism is its adaptability and resilience in response to a swiftly evolving society. Traditional East Asian authoritarian developmentalism, a strategy for political survival, has often been viewed as inherently short-lived due to mounting societal pressures for democratic reform (Castells 1992). Yet the resilience of these states plays a crucial role in the contemporary paradigm of East Asian authoritarian developmentalism, enabling them to withstand societal demands for change in the digital age. Central to this resilience is a strategy of authoritarian adaptation. As economic development progresses, the developmental state undergoes transformation, driven by the burgeoning middle class's advocacy for democratic transition (Lipset 1959). In cases where such transformation does not occur, authoritarian developmental regimes must adapt strategies of political centralization. Amid a rapidly evolving society, these regimes often employ authoritarian co-optation to address the aspirations of the emerging middle class (He 2021).

This article emphasizes the techno-developmental state's use of digital technologies for governance, a crucial aspect of authoritarian adaptation in the digital age to address middle-class aspirations. The integration of digital technologies not only strengthens existing authoritarian structures but also equips these resilient regimes with additional tools from emerging technologies to enhance their capacity to deliver public services and ensure political longevity. This trend is particularly evident in the persistent authoritarian developmental states of East Asia. In Singapore, policymakers have actively addressed governance challenges by embracing technologies such as data analytics (Woo 2018). Similarly, in the context of China's "developmental

authoritarianism” (Yang 2017), digital governance enhances the state’s ability to tackle social grievances, enhance public service delivery, and promote social stability (Hu and Zhang 2023; Lin, Wang, and Tsai 2022; Zeng 2020).

Conceptualization of the Techno-Developmental State and Research Design

This body of literature provides a theoretical framework for understanding the techno-developmental state. Table 1 outlines the distinctions between the traditional developmental state model and its evolved form, the techno-developmental state. This evolution occurs against the backdrop of the digital era and is significantly influenced by the dynamics of great power politics.

The trajectory of great power politics underscores a heightened emphasis on technology, illustrated by the rise of governmental bodies overseeing science and technology and the state’s alignment with the technology sector. Consequently, the techno-developmental state functions to address both internal imperatives for legitimacy and external challenges posed by great power dynamics.

The digital era impacts the state in two primary ways: by prioritizing technological advancements to drive economic development (Lei 2023); and by using technologies to strengthen social governance. Ultimately, leveraging

TABLE 1. Comparison of the Developmental State and the Techno-Developmental State

	<i>Developmental state</i>	<i>Techno-developmental state</i>
<i>Key motives</i>	Regime survival	Roots in regime survival A tech focus intensified by great power politics
<i>Leading government agencies</i>	Economic planning agencies	Economic planning agencies Agencies responsible for science and technology
<i>State–business relations</i>	State–business collaboration for economic growth	State–business collaboration for growth with the state’s realignment with tech firms in strategic areas
<i>Economic governance</i>	Formulating industrial policy to transform the economy	Strategic policy to harness digital technologies to transform the economy
<i>Social governance</i>	Authoritarian control	Authoritarian co-optation enabled by emerging technologies

enhanced technological capabilities, the state advances its developmental agenda through strategic initiatives aimed at harnessing technology for economic growth and effective social governance in a new era of authoritarian developmentalism.

This article uses the Chinese state as a case study to explore the emergence and operation of the techno-developmental state, as outlined in Table 1. It aims to elucidate the rise of new East Asian authoritarian developmentalism in the digital era amid escalating great power politics. The upcoming section addresses the evolution of the Chinese developmental state and its adaptation of social governance strategies in response to the digital era, as well as the impact of heightened US–China technological competition, which has led to the establishment of government agencies overseeing science and technology and the recalibration of state–business relations under a new technological focus.

The next section analyzes the efforts of the Chinese techno-developmental state to advance economic and social governance objectives in the early 2020s. This section provides an extensive analysis of the New Infrastructure initiative, through which the Chinese state actively promotes strategic emerging technologies such as 5G, artificial intelligence, the industrial internet, and the Internet of Things. This initiative was introduced during the Central Economic Work Conference in 2018 and underscored in the 2019 Government Work Report at the National Two Sessions (*CPC News* 2020), coinciding with heightened US–China tech competition. This case study illustrates how the techno-developmental state uses emerging technologies to achieve economic development objectives and enhance social governance mechanisms.

THE RISE OF THE TECHNO-DEVELOPMENTAL STATE IN CHINA

Regime Survival and the Persistence of Performance Legitimacy in a Digital Era

The genesis of the Chinese techno-developmental state can be traced back to two significant political upheavals that prompted the state to pursue performance-based legitimacy. Following Mao Zedong's death in the late 1970s, the Communist Party's political authority recognized that it needed to transition from relying solely on political mobilization and control to achieving legitimacy through performance (Harding 1987; Shirk 1993; White 1993). The Chinese state therefore began to seek performance-based

legitimacy predominantly through market reforms (Harding 1987; Shirk 1993; Zhang 2000), leading to the establishment of “market-compatible institutions” in the post-Mao era (Qian 2017).

The imperative for performance legitimacy was further emphasized following the Tiananmen crisis of 1989, prompting an intensified iteration of Deng’s original model of economic reform and opening-up (Lieberthal 1995; Naughton 2006; Zhang 2000). In pursuit of a formula for performance legitimacy, a series of economic agencies have played pivotal roles in China’s market reform and economic restructuring, culminating in the establishment of the influential National Development and Reform Commission (NDRC) in 2006. The statist approach to economic development has persisted in recent decades alongside efforts to strengthen governance mechanisms through internal reforms. Under Hu Jintao’s leadership, the state promoted the 2006 Indignous Innovation campaign and maintained a state–society relationship through systematic political repression (Fewsmith 2008; Pei 2006; Yang 2004). This statist economic strategy, orchestrated by an increasingly centralized illiberal state, has become a defining feature of Xi Jinping’s leadership (Economy 2018).

While the foundational principles of the developmental state model have remained unchanged, the advent of digital technologies has significantly strengthened the authoritarian structure’s capacity for political control. Central to this evolution is the expansion of China’s surveillance apparatus, which has been a cornerstone of the state’s authoritarian framework since the end of the Mao era (Pei 2024). The arsenal now includes extensive mass video-surveillance initiatives incorporating facial-recognition technology, voice-recognition software, and a comprehensive DNA collection program. These initiatives are complemented by the nationwide Social Credit System, designed to assess the behavior of every Chinese citizen (Cabestan 2020; Qiang 2019). Surveillance technologies originally intended to support more “service-oriented” governance in China’s “smart city” projects are now perceived as intrusive tools of political control (Arcesati 2022; Hoffman 2022).

To strengthen its surveillance capabilities in the digital age, the state has partnered with leading domestic private firms to adopt foreign technologies and drive innovations in digital surveillance (Huang and Tsai 2022). This robust techno-authoritarian initiative aims to address and control threats to public health and public order. The onset of COVID-19 further bolstered this effort, prompting the state to implement an illiberal and autocratic model in its response to the pandemic (Greitens 2020).

Institutional Rebuilding for Technological Development amid the US–China Technology Rivalry

The intensifying US–China technology war since 2018 has played a pivotal role in catalyzing the emergence of the techno-developmental state. Since 2016, Chinese leaders have championed the “implementation of the strategy of innovation-driven development,” underscoring the nation’s ambitious aims in global high-tech competition to realize the Chinese dream of national rejuvenation and global prominence by midcentury (Cheung 2022). This endeavor gained significant momentum amid the escalating US–China technology rivalry of the late 2010s.

The US sanctions against Chinese tech companies prompted the Chinese government to prioritize a New Whole-State System. This system has been prominently featured in policy reports delivered by Xi Jinping, including those presented at key Party and governmental forums such as the 20th National Party Congress, the 14th Five-Year Plan (for 2021–2025), and Long-Range Objectives for the Year 2035 (Zhang and Lan 2023). The formulation of techno-economic policies aimed at fostering science and technology innovation has since become integral to the Party and national development agenda. Significantly, in December 2021, science and technology policy was included for the first time in the seven major policies discussed at the Central Economic Work Conference in Beijing (Yuan 2022).

The increasing demand for techno-economic policy necessitated two significant rounds of statist institutional rebuilding, in 2018 and 2023, respectively. In 2018, the state restructured its institutional R&D framework by integrating the State Administration of Foreign Experts Affairs into the Ministry of Science and Technology (MST) to optimize the utilization of international high-tech talent. The MST also took over management of the National Natural Science Foundation of China (NSFC) to tighten oversight of national R&D programs aligned with the Party’s emphasis on high-tech development, especially in semiconductors and AI. This restructuring aimed to enhance fund allocation toward nationally prioritized science and technological projects (Cheng 2023).

Further statist institutional rebuilding occurred in response to new US export controls in 2022 and heightened the urgency of bolstering resilience to external shocks. China’s techno-economic policy shifted its focus to “bottle-neck” (*qia bozi*) issues in strategic emerging technologies, emphasizing the

promotion of basic scientific research and improvement of infrastructure for nationally prioritized projects. The state's immediate response included appointing five prominent scientists to the 24-member Politburo, historically dominated by economic technocrats, at the Party's 20th National Congress (in October 2022) to ensure the country's technological self-reliance (Yu 2023).

In March 2023, the state's intensified focus on technology prompted further restructuring within the MST. This round of institutional rebuilding witnessed the establishment of the Central Science and Technology Commission, which transferred responsibility for advanced scientific and technological R&D to the Politburo, with a focus on consolidating national resources to address technical challenges, particularly in advanced chip R&D (Cheng 2023). The restructured MST launched the Artificial Intelligence for Science project in collaboration with the NSFC, signaling the state's renewed commitment to addressing bottlenecks across strategic emerging sectors (*Global Times* 2023). The latest reorganization of the MST is also expected to enhance the role of the private sector within the New Whole-State System, which is crucial for China's state-led technological advancement (Cheng 2023).

The Mobilization of the Tech Sector and the Remaking of the State–Business Alliance

The state's drive to enhance technological self-sufficiency in strategic sectors amid the escalating US–China tech competition calls for the mobilization of the entire domestic tech sector. This mobilization has primarily involved two categories of firms. First, state-owned enterprises (SOEs) were identified by the government as pivotal for technological autonomy in the face of increased US tech restrictions (Reuters 2023). In 2021, the State-Owned Assets Supervision and Administration Commission released a list highlighting 178 technology innovations in critical areas by centrally administered SOEs (Li 2021). By 2023, the list had grown to 369 innovations.¹

Second, intensified external pressures have spurred the mobilization of small and medium-sized enterprises (SMEs) in strategic technological sectors to support the state's economic objectives. Drawing inspiration from Germany's

1. The list is available at <http://www.sasac.gov.cn/n4470048/n22624391/n26705666/>.

“hidden champions”—highly specialized SMEs dominating niche global markets—China recalibrated its economic strategy to foster high-tech SMEs (Brown, Chimits, and Sebastian 2023). This strategic focus on SMEs is exemplified by China’s Little Giants Initiative, a program to support startups positioned to fill technology gaps in critical sectors. Between 2019 and 2022, the Ministry of Industry and Information Technology (MIIT) identified 4,762 “little giants,” many specializing in semiconductors, machinery, and pharmaceuticals (Bloomberg News 2022). This initiative provides extensive support to eligible SMEs through the innovation chain model and facilitates their access to financial markets, including the Beijing Stock Exchange, which was established in 2021 to bolster China’s tech SMEs (Bloomberg 2021).

The expansion of the state’s technological ambitions necessitated a recalibration of its relationship with domestic tech giants, primarily in the consumer technology sector, as it concentrated financial resources in strategic tech sectors. Between late 2020 and early 2022, a sweeping regulatory crackdown led to significant restructuring of major tech firms to reduce their influence in the consumer tech sector and redirect their focus toward strategic domains (Lei 2023, 276–92; Zhang 2024). Regulatory efforts particularly targeted what Liu (2021) identifies as the “steering object”—China’s attempt to curb the influence of big fintech platforms, notably Alibaba’s Alipay and ANT Financial.

By January 2022, the state realized that aligning with these tech firms was crucial for achieving economic growth and the state’s technological ambitions. Consequently, the state directed regulators to support big tech firms in advancing key software technology research in industrial internet and blockchain. This directive also encouraged increased R&D investment to accelerate technological breakthroughs in critical areas such as AI, cloud computing, and blockchain (US Chamber of Commerce 2022, 45). This effort was furthered in July 2023 when premier Li Qiang held discussions with major platform companies such as Meituan, Alibaba Cloud, and Douyin, following the NDRC’s endorsement of investment projects by leading platform companies (Tang and Nulimaimaiti 2023).

The final aspect of reshaping state–business relations involves the state’s recalibration of its interactions with foreign tech firms. While the state’s regulatory measures were not explicitly targeted at these firms, they significantly impacted its engagement with multinational corporations, leading to the withdrawal of several tech firms from the Chinese market (Zhang 2024, 205). These regulations arguably strengthened the state’s bargaining power

with foreign tech firms (Klyman 2022), facilitating a swift realignment to attract essential foreign investment for economic growth. This was exemplified in April 2024 when Tesla collaborated with Baidu to introduce its semi-autonomous driving technology in China, navigating regulatory challenges through discussions with premier Li Qiang (White and Campbell 2024).

Using Emerging Technologies to Facilitate Social Governance

The rise of the techno-developmental state is also driven by the domestic imperative to use digital technologies to enhance capabilities in social governance, to reinforce the state's authoritarian structure. The focus on digital technologies primarily aims to improve the efficiency of public service delivery. Throughout the 2010s, there was a concerted effort to improve public services by promoting interconnectedness and collaboration among government departments, addressing the challenge of "information silos" (Lin, Wang, and Tsai 2022). Two key initiatives were implemented: the establishment of a digital service system; and the open sharing of government data.

The state has recognized the transformative role of digital technologies in governance services, to reduce information barriers and silos between government units. A significant part of this approach was the Internet Plus strategy, which was launched as part of the 13th Five-Year Plan in 2016 and aimed to establish the first "state services system." Then-premier Li Keqiang underscored the internet's potential to enhance public service accessibility and improve government administration (*Govinsider* 2016). And techno-economic policy initiatives such as the Outline of Operations to Stimulate the Development of Big Data (2015) and the New Generation Artificial Intelligence Development Plan (2017) highlighted the critical role of harnessing emerging digital technologies in advancing public services.

Another crucial aspect addressed in these documents was the initiative to accelerate the open sharing of government data (State Council 2015). Substantial advancements were achieved by 2019, significantly enhancing the openness and accessibility of government data. This effort allowed both government departments and the general public to access a wider array of valuable data and information resources (Central Government 2019). This progress also led to increased government investments in big data, contributing to a rapid expansion of the big data governance market, which grew by a notable 31% in 2019 (*Sohu* 2020).

In the early 2020s, with the state's newly gained tech focus shaped by intensifying great power politics, the Chinese state upped its efforts to integrate digital technologies with social governance, as exemplified by the inclusion of the “digital government” concept in the 14th Five-Year Plan, issued in March 2021. This plan outlined three key objectives: enhancing the openness and sharing of public data; promoting the collaborative development and utilization of digital government services; and improving the efficiency of digital government services (NDRC 2022a). In June 2022, the State Council issued the Guiding Opinions on Strengthening the Construction of Digital Government, emphasizing the establishment of a framework aligned with modernizing the national governance system and improving the efficiency of public service provision (Central Government 2022). The “digital China” initiative outlined in the report of the 20th National Congress further underscored the importance of modernizing the social governance system through the establishment of a digital government.

This heightened commitment subsequently intensified efforts to integrate digital technologies into social governance, as reflected in two key policy documents. The National Informatization Plan for the 14th Five-Year Plan, revealed in December 2021, aims to establish digital social governance by 2025, enhancing government service through extensive big data adoption across all levels (Cyberspace Administration 2022). The Overall Layout Plan for Building a Digital China, unveiled in February 2023, introduces “digital governance” (*shuzi zhengwu*) for the first time, reflecting dual state intentions: promoting targeted digital technology use in national governance, and fostering links between digital transformation in social governance and other sectors (Gao and Li 2023).

THE TECHNO-DEVELOPMENTAL STATE IN ACTION: THE NEW INFRASTRUCTURE INITIATIVE

In the early 2020s, the Chinese techno-developmental state launched a substantial project with the dual objectives of economic growth and effective social governance through the strategic use of digital technologies. A cornerstone of this project was the New Infrastructure initiative, through which the state actively pursued a tech-led growth agenda and bolstered its technology-enabled social governance mechanisms.

The Digital Transformation of China's Industrial Base

The techno-developmental state's pursuit of a tech-led growth agenda has been driven by its efforts to enhance economic prospects. During the COVID-19 pandemic, China aimed to generate employment opportunities and stimulate economic expansion without overly relying on traditional stimulus measures (Meinhardt 2020). Recognizing the constraints of investing in “old infrastructure” and learning from the 2009 financial crisis, investing in “new infrastructure” emerged as a more promising strategy due to its potential to substantially bolster economic growth and promote innovation-driven development (Ye 2020).

In the longer term, the development of digital infrastructure is poised to become a pivotal driver of industrial transformation. This recognition was underscored by the government during an executive meeting of the State Council in April 2020, acknowledging digital infrastructure as a key economic multiplier essential for advancing industrial upgrading across sectors (Central Government 2020). The structural transformation of the economy through digital technologies gained momentum as a response to enhancing economic resilience amid the challenges of COVID-19. Emphasizing the advantages of new infrastructure, such as its shorter iteration cycle, enhanced resilience, and flexibility in responding to unforeseen events, the government prioritized the robust expansion of the digital economy as a cornerstone of economic resilience (Xu 2022). Consequently, in October 2022, two influential technocrats who had been pivotal in advancing provincial digital economy growth were appointed to the Politburo (Yu 2023, 19).

A major focus of the state's pivotal digital transformation initiative is the revitalization of the country's heavy industries. In November 2019, the MIIT launched an initiative called the Promotion Plan for the 5G+ Industrial Internet 512 Project. It set broad guidelines, and from April 2020 to November 2021, the government actively facilitated the integration of 5G+ industrial internet technology across various major traditional manufacturing sectors (Table 2). The state also mobilized central SOEs to advance the construction of the industrial internet. By 2023, 71 central SOEs had established more than 150 high-level industrial internet platforms, with a networking rate over 50% for key equipment (Huawei 2023).

The 2022 Evaluation Report on the Development Effectiveness of China's 5G+ Industrial Internet, published at the end of 2022, highlights the success

TABLE 2. Key Steps in the Implementation of the 5G+ Industrial Internet Initiative

<i>Date</i>	<i>Policy document</i>	<i>Affected industries</i>
November 2019	Promotion Plan for the 5G+ Industrial Internet 512 Project	Called for the selection of 10 key industries
April 2020	Guidelines for Intelligent Mine Construction in the Nonferrous Metal Industry (Trial)	Nonferrous metals
April 2020	Guidelines for the Construction of Intelligent Smelting Plants in the Nonferrous Metal Industry (Trial)	Nonferrous metals
May 2020	Guidelines for the Construction of Intelligent Processing Factories in the Nonferrous Metal Industry (Trial)	Nonferrous metals
September 2020	Building Materials Industry Intelligent Manufacturing Digital Transformation Action Plan (2021-2023)	Building materials
May 2021	Ten Typical Application Scenarios and Five Key Industry Practices of 5G+ Industrial Internet	Electronic equipment manufacturing, equipment manufacturing, steel, mining, and power industries
November 2021	Ten Typical Application Scenarios and Five Key Industry Practices of 5G+ Industrial Internet (Second Batch)	Petrochemicals, building materials, port, textile, and home appliances

Source: Compiled by the author based on CAICT (2021) and the listed policy documents.

of the state’s policy initiatives in preceding years. According to the report, more than 4,000 “5G+ industrial internet” projects have been implemented across 41 major economic sectors nationwide (Alliance of Industrial Internet 2022). Following this assessment, a new series of government initiatives was launched to further advance the integration of 5G technology and the industrial internet. In July 2023, the MIIT introduced the 2023 Work Plan for the Industrial Internet Special Working Group to enhance the manufacturing sector through the promotion of a new phase of industrial internet development. The plan is designed to facilitate the establishment of approximately 3,000 5G-enabled factories within a year. It also promises to advance upgraded versions of “5G+ industrial internet” projects (Ma 2023).

The structural transformation of the economy will also mean a digital revolution within the SME sector, which has long been constrained by significant financial barriers to the adoption of digital technologies. Despite SMEs comprising over 90% of the country's manufacturing industry, a substantial portion—more than 55% as of 2020—struggled with basic equipment digitalization (Yu 2020). The economic repercussions of the COVID-19 pandemic further emphasized the urgency of addressing SMEs' challenges, prompting the government to prioritize the promotion of digital transformation among these enterprises. Recognizing the limited digital maturity of SMEs, in April 2020 the MIIT issued a Special Action Plan on the Digital Empowerment of SMEs. This comprehensive plan included a diverse range of policy measures such as inclusive financing and ecosystem development (CCTV 2020).

As the country transitions toward an increasingly digital economy, the development of computing power infrastructure is crucial for transforming the industrial base. China formally established the National Data Administration (NDA) in late 2023 as a dedicated national bureau under the NDRC tasked with advancing datacentric economic development and overseeing the market (State Council 2023). The NDA's key objectives include standardizing industry data norms, promoting cross-sector data sharing, and harmonizing data regulations across various government bodies. The head of the NDA soon introduced the “Data Element ×” initiative to harness data's multiplier effects across all sectors, with the overarching goal of “building a digital China” (NDRC 2023).

One obstacle to a digital transformation in the early 2020s is the uneven distribution of computing power across China. As of 2021, internet data center resources in Beijing, Shanghai, and Guangzhou-Shenzhen collectively accounted for 65.3% of the national share, with Beijing at 23.3%, Shanghai at 16.4%, and Guangzhou-Shenzhen at 13.7%. In contrast, the central, western, and northeastern regions combined represented less than 30% (Liu 2022). To address this disparity, a significant initiative known as Eastern Data and Western Computing has been underway since May 2021. By February 2022, the NDRC and two other departments had approved multiple projects, including the construction of national computing hub nodes in the Beijing-Tianjin-Hebei area, the Yangtze River Delta, and the Guangdong-Hong Kong-Macao Greater Bay Area, as well as 10 national data center clusters (NDRC 2022b). This initiative gained momentum in

late 2023 with the release of an implementation plan aiming to establish a comprehensive computing power infrastructure system by the end of 2025 (*Xinhua* 2023).

Upgrading the Social Governance Mechanism

The New Infrastructure initiative also provided the techno-developmental state with an opportunity to enhance its capacity for authoritarian governance. The foundation of the government cloud can be traced back to the “cloud computing service priority” strategy outlined in the National E-Government 12th Five-Year Plan of 2012, a significant initiative to improve governance efficiency through technological advancements. To strengthen its capabilities in social governance, the state has focused on the development of integrated social governance systems harnessing various digital technologies. This objective was underscored by the NDRC’s 14th Five-Year Plan for Advancing National Government Informatization, published in December 2021. The plan highlighted the imperative to establish “integrated and intelligent government cloud platform systems” with robust computing power and advanced algorithmic support capabilities. These systems integrate cutting-edge technologies such as big data and AI (Huawei and CAICT, 2023).

In 2022 the government embarked on a project to advance its social governance system. Part of this project is an ongoing effort to enhance cloud governance by establishing a nationwide integrated government affairs big data system. It sought to integrate the national government, provincial governments, and relevant agencies by the end of 2023, with further improvements slated for 2025 (State Council 2022). The initiative, articulated in the White Paper on Integrated and Intelligent Government Cloud Construction and Operation released in February 2022, was designed to address identified deficiencies in government cloud infrastructure, including decentralized management of cloud resources, inadequate support for unified browsing and application, and the absence of a unified platform capable of coordinating cloud resources (Xie 2022).

The growth of the cloud governance sector has played a pivotal role in advancing the government’s modernization of social governance systems. Prior to the various new digital governance initiatives of the early 2020s, this sector was already experiencing rapid expansion. In 2017, propelled by the

vigorous promotion of “internet + government services,” the government cloud market surged to over 27 billion yuan, with a remarkable year-on-year growth of 41.2% (*The Paper* 2020). By 2018, the adoption of cloud services in governmental affairs had consistently surpassed sectors such as industry, finance, and the internet (OSCAR 2018).

This growth trend continued into the early 2020s. According to a report by market research firm IDC, China’s government cloud market surpassed 50 billion yuan in 2022, marking a 17% year-on-year increase (CIIA 2023). During this period, Huawei actively participated in over 800 domestic government cloud projects and established government cloud operation centers in more than 150 cities nationwide (*Huawei* 2023). Leveraging advanced technologies such as generative AI, Huawei launched the National Cloud 2.0 in February 2024, to “accelerat[e] the intelligent transformation of public services to drive cognitive society” (PR Newswire 2024). Huawei, Alibaba Cloud, and Inspur Cloud have emerged as key players in the government’s initiative to build integrated big data governance systems (IDC 2023).

Alongside the state’s efforts to enhance its cloud governance system, significant strides have been made in developing e-government networks overseen by the State Information Center (Administration Center 2021). The heightened demand for effective social governance amid the challenges posed by COVID-19 underscored the critical role of e-government networks in the state’s social governance agendas. In October 2020, for example, the State Information Center organized a National Government External Network CIO Advanced Training Class in Guiyang, focusing on “maximizing the role of government external networks in epidemic prevention and control” (NDRC 2020).

To enhance the state’s capacity for social governance, the primary objective of upgrading e-government networks is to improve connectivity and efficiency across government units through the use of digital technologies. Several relevant initiatives have been undertaken. First, in July 2021, the central government initiated a project to transition national e-government external networks, local government external networks, and government-dedicated networks to Internet Protocol version 6 (Central Government 2021). This initiative was driven by the state’s goal to enhance governance efficiency in response to the growing demand for interdepartmental collaboration (Administration Center 2021). Second, in 2021, the state standardized the construction of the mobile end of the integrated national government service

platform to facilitate interconnectivity among government service platforms across regions and departments (*Xinhua* 2021a). The state has also recognized the importance of 5G technology for enhancing government efficiency. China's first provincial-level 5G electronic government external network was launched in Guangdong in November 2021 to enhance the efficiency of public services by enabling provincial public servants to access e-government networks (*Xinhua* 2021b).

Statistics provided by the State Information Center indicate the robustness of the cross-departmental and cross-regional social governance system as of April 2024. The government's external network now spans all administrative regions at the central, provincial, municipal, and county levels. It connects 225 central departments and related agencies, as well as 518,500 local government departments and related agencies. The access rate of townships and streets has surpassed 99.4%. Government clouds have been established in 31 provinces (regions and municipalities), and government cloud services have been implemented in 95% of prefecture-level cities (NDRC 2024).

The primary aim of the government's digital initiatives has been to streamline public service delivery. Alongside efforts to enhance the efficiency of the e-government system, there has been a concurrent focus on meeting societal demands through improved public service provision. The National Integrated Government Services Enhancement Project was highlighted in the 14th Five-Year National Informationization Plan of 2021, advocating the establishment of a performance review system that allows the public to assess public service delivery. In 2023, this initiative was further strengthened through the introduction of a "regularized work mechanism" (*changtaihua gongzuo jizhi*) aimed at optimizing government service effectiveness and leveraging upgraded e-government networks (Central Government 2023).

CONCLUSION

The article has explored the emergence of the techno-developmental state using China as an empirical case study. I argue that the rise of the Chinese techno-developmental state stems from two strategic maneuvers of authoritarian developmental states: enhancing their influence over technological sectors amid escalating US–China technology competition; and leveraging digital technologies to bolster social governance in the digital era. To investigate the techno-developmental state's strategic actions in pursuing its

economic and social governance agendas, we considered the New Infrastructure initiative. I believe this analysis can also be applied to other initiatives, such as the promotion of “smart cities,” where Chinese authorities have accelerated the adoption of technologies to pursue economic growth opportunities while also strengthening social governance mechanisms.

This analysis aims to deepen our understanding of authoritarian developmentalism in China, underscored by the state’s strategic cultivation of technologies in the context of the digital era and escalating global power dynamics. However, while the Chinese experience provides a valuable empirical basis for understanding how authoritarian developmental states adapt in this new era, the rise of the techno-developmental state extends beyond China. Singapore, like China, has shown resilience in maintaining authoritarian developmentalism amid societal pressures by systematically leveraging technologies. As a small state navigating the growing regional geoeconomic challenges, Singapore has been driven to protect its technological edge amid heightened US–China technological competition. I believe that my argument can also shed light on Singapore’s security-focused, technology-driven authoritarian developmentalism.

Moreover, it is crucial to consider the broader implications, beyond its borders, of the techno-developmental state’s rise in China. Studies indicate that countries in the Global South are increasingly adopting Chinese e-government technologies (Arcesati 2022). In the healthcare sector, Chinese firms have exported technological solutions for “epidemic prevention and control” globally. For instance, during the pandemic, Pakistan adopted and adapted China’s effective e-governance model through initiatives like the China–Pakistan Economic Corridor (Ullah et al. 2021). Future research should delve deeper into the international dimensions of the techno-developmental state’s rise by examining the specific mechanisms deployed by the Chinese state to extend its global economic influence. This exploration could shed light on whether this phenomenon might catalyze the emergence of techno-developmental states in other countries.

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